

We have two types of user in cloudera

- **Normal user ()**

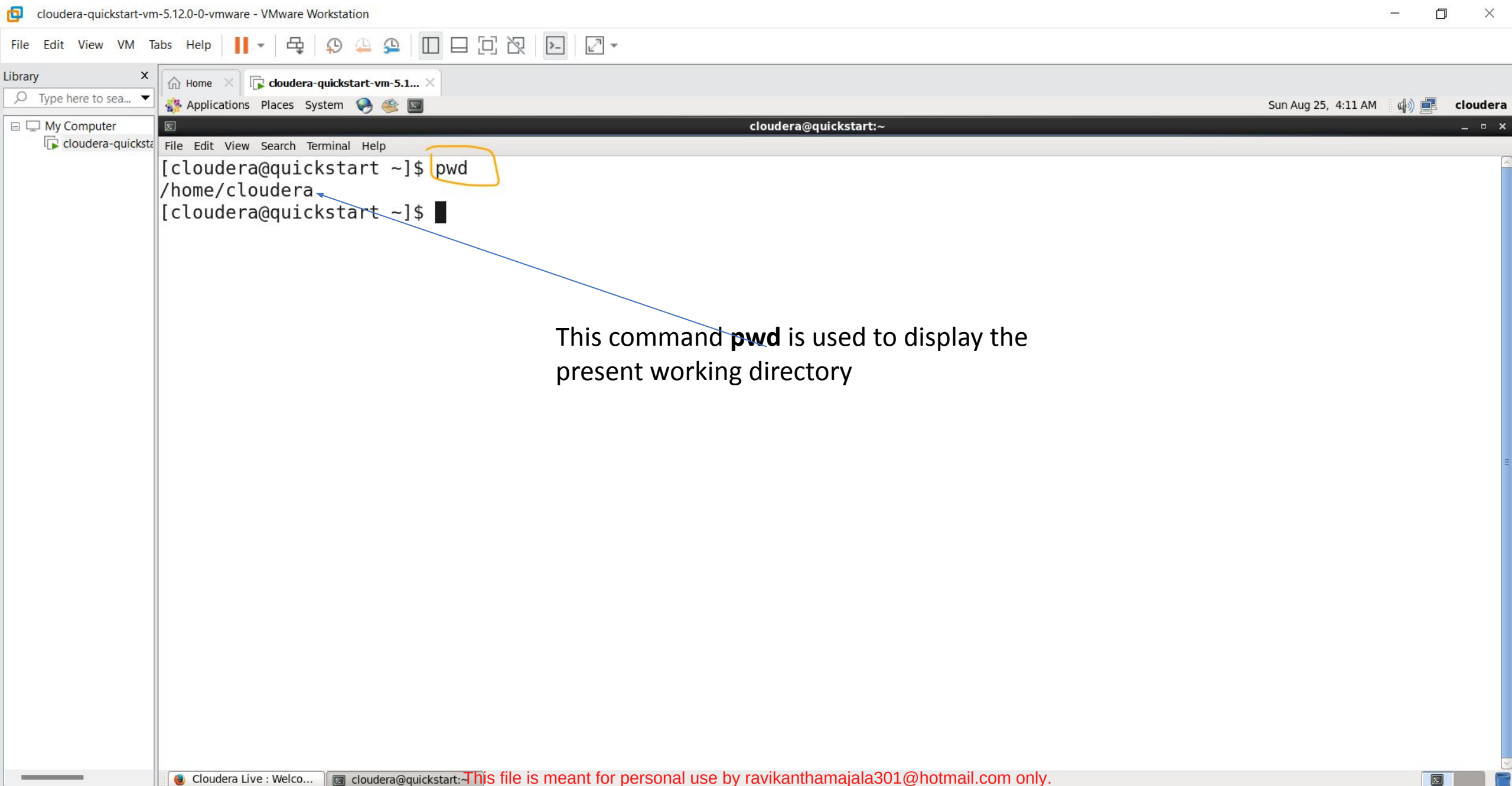
cloudera /cloudera

- **Root user**

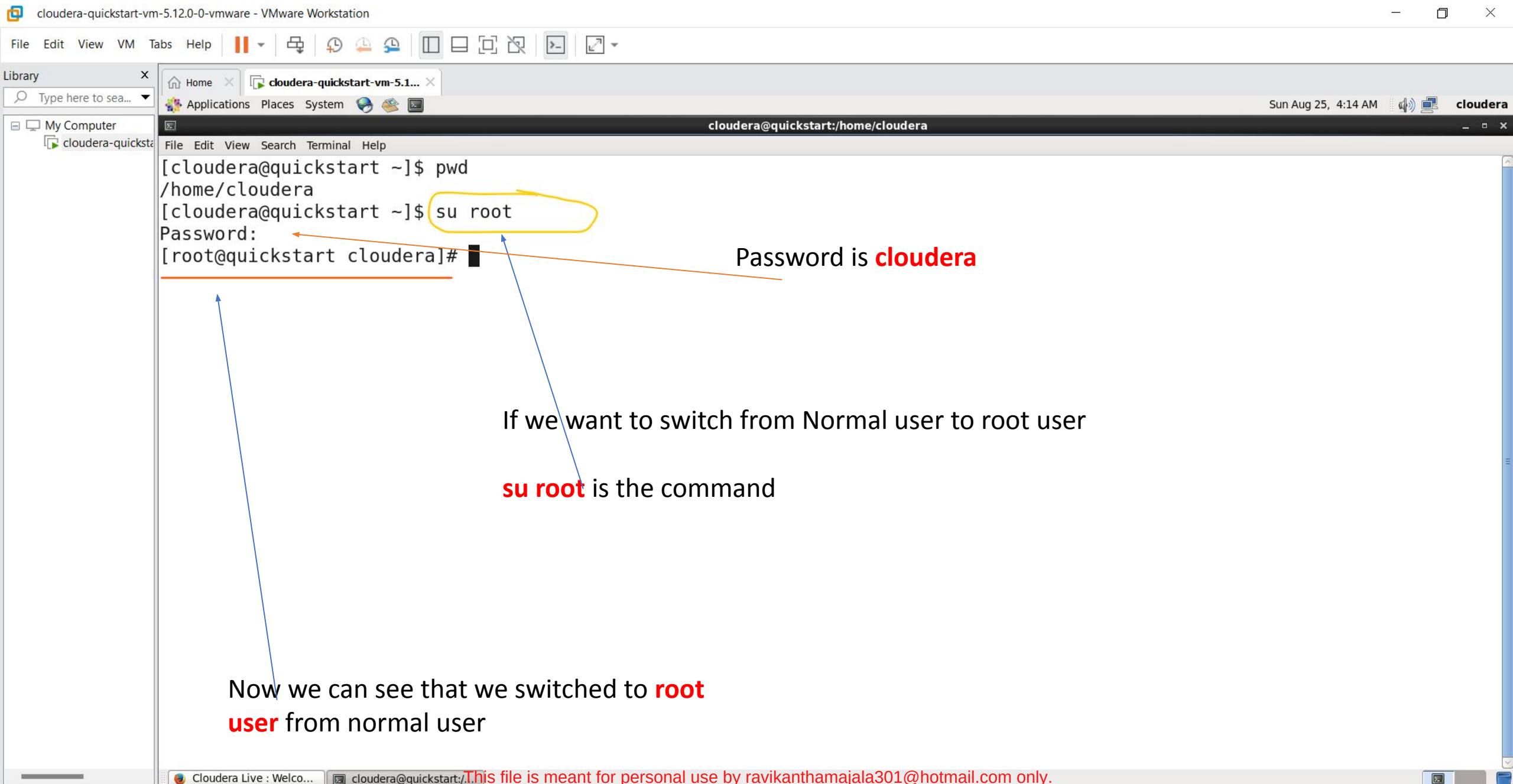
- The **root user** has the highest level of privileges in the system.
- It can install software, modify system files, manage services, and control other users.

User name :Root

Password :cloudera



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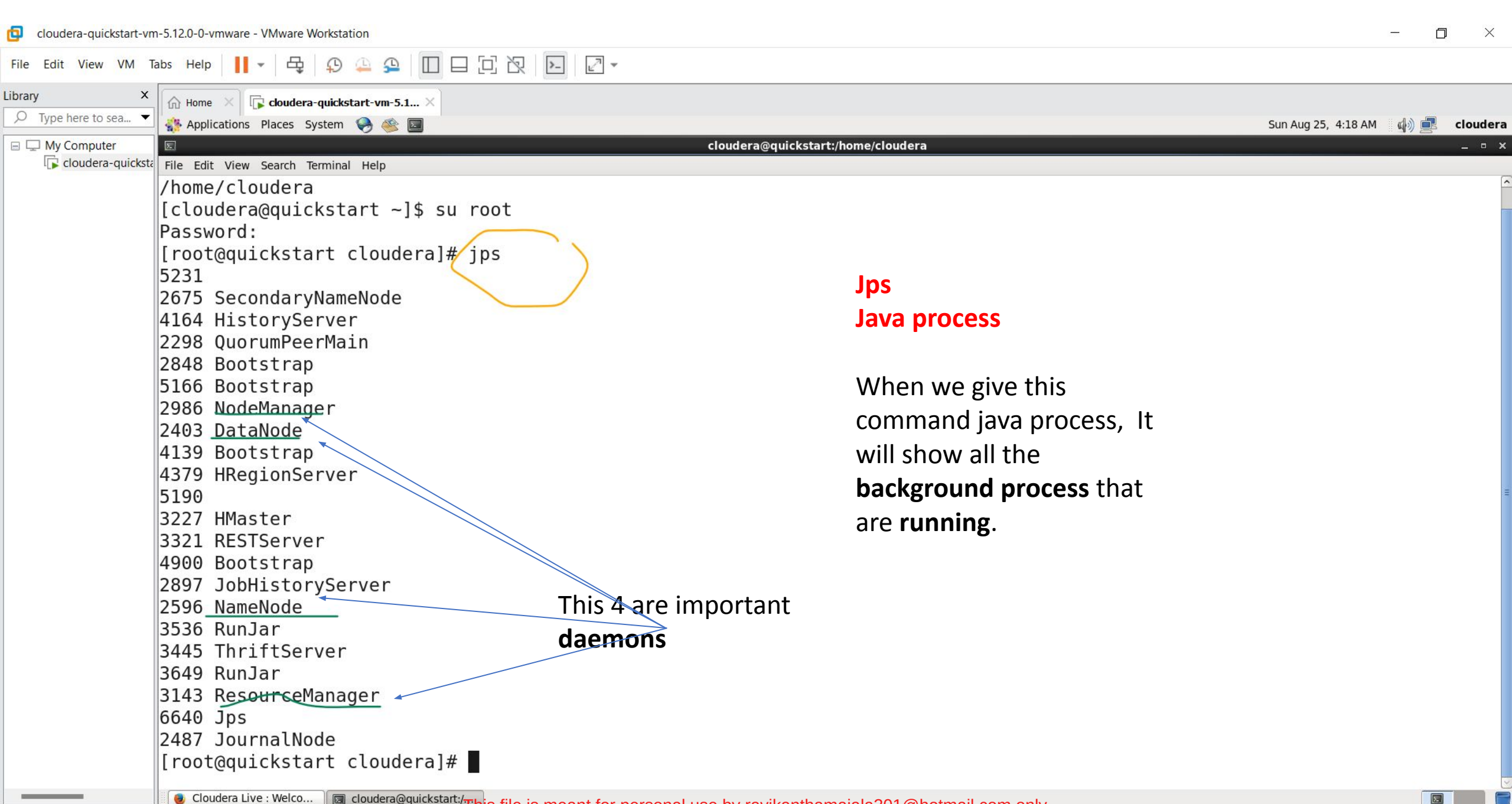
```
[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$ su root
Password:
[root@quickstart cloudera]#
```

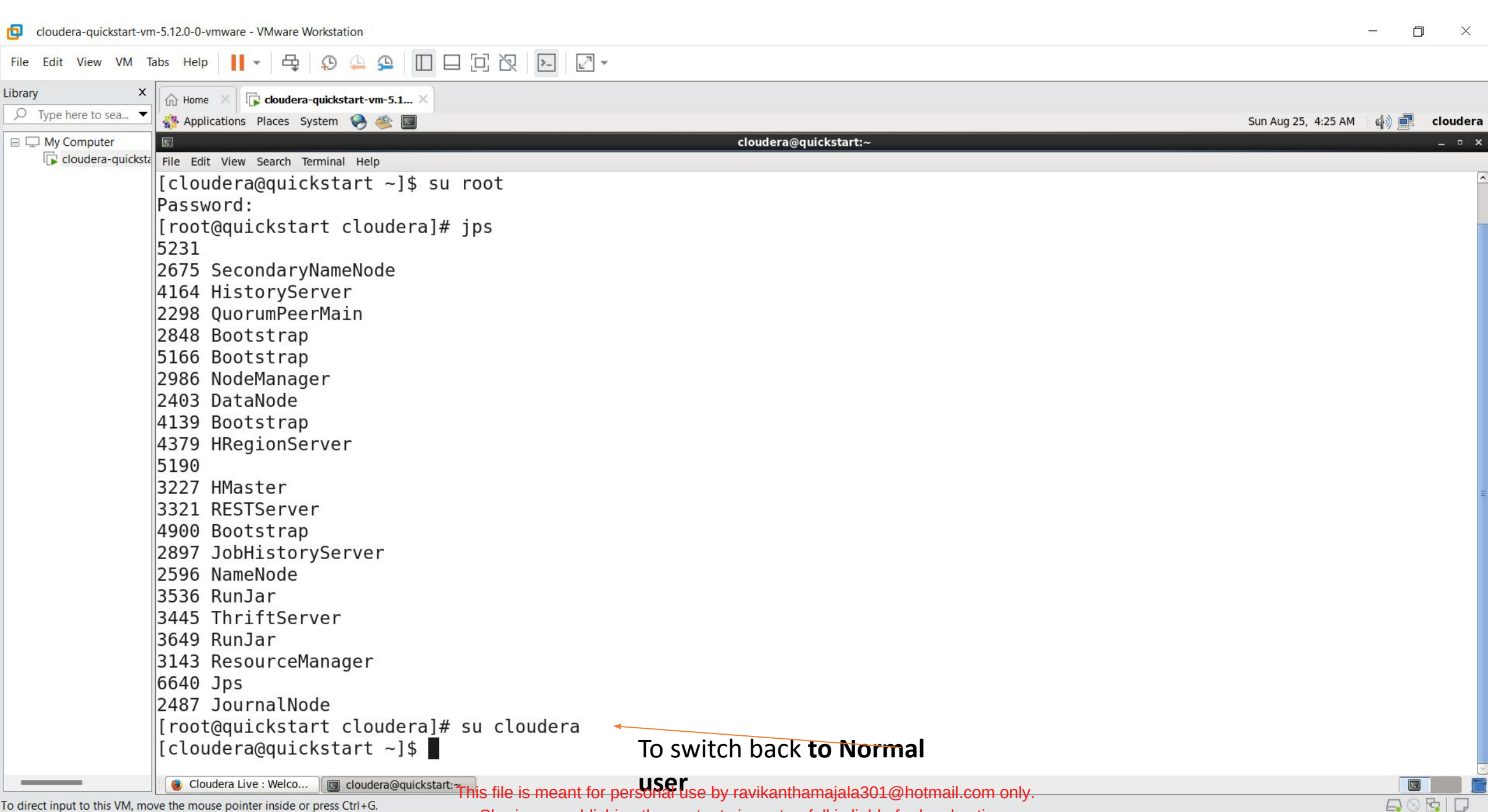
Password is **cloudera**

If we want to switch from Normal user to root user

su root is the command

Now we can see that we switched to **root user** from normal user





To switch back to Normal
user

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```
[cloudera@quickstart ~]$ hdfs fsck /
```

```
Connecting to namenode via http://quickstart.cloudera:50070/fsck?ugi=cloudera&path=%2F
```

```
FSCK started by cloudera (auth:SIMPLE) from /127.0.0.1 for path / at Sun Aug 25 04:29:34 PDT 2024
```

[illegible]

✓ Status: HEALTHY

Total size: 858124212 B (Total open files size: 166 B)

```
Total dirs: 89
```

Total files: 934

```
Total symlinks: 0 (Files currently being written: 3)
```

Total blocks (validated): 931 (avg. block size 921723 B) (Total open file blocks (not validated): 2)

Minimally replicated blocks: 931 (100.0 %)

```
Over-replicated blocks:      0 (0.0 %)
```

Under-replicated blocks: 0 (0.0 %)

Mis-replicated blocks: 0 (0.0 %)

Default replication factor: 1

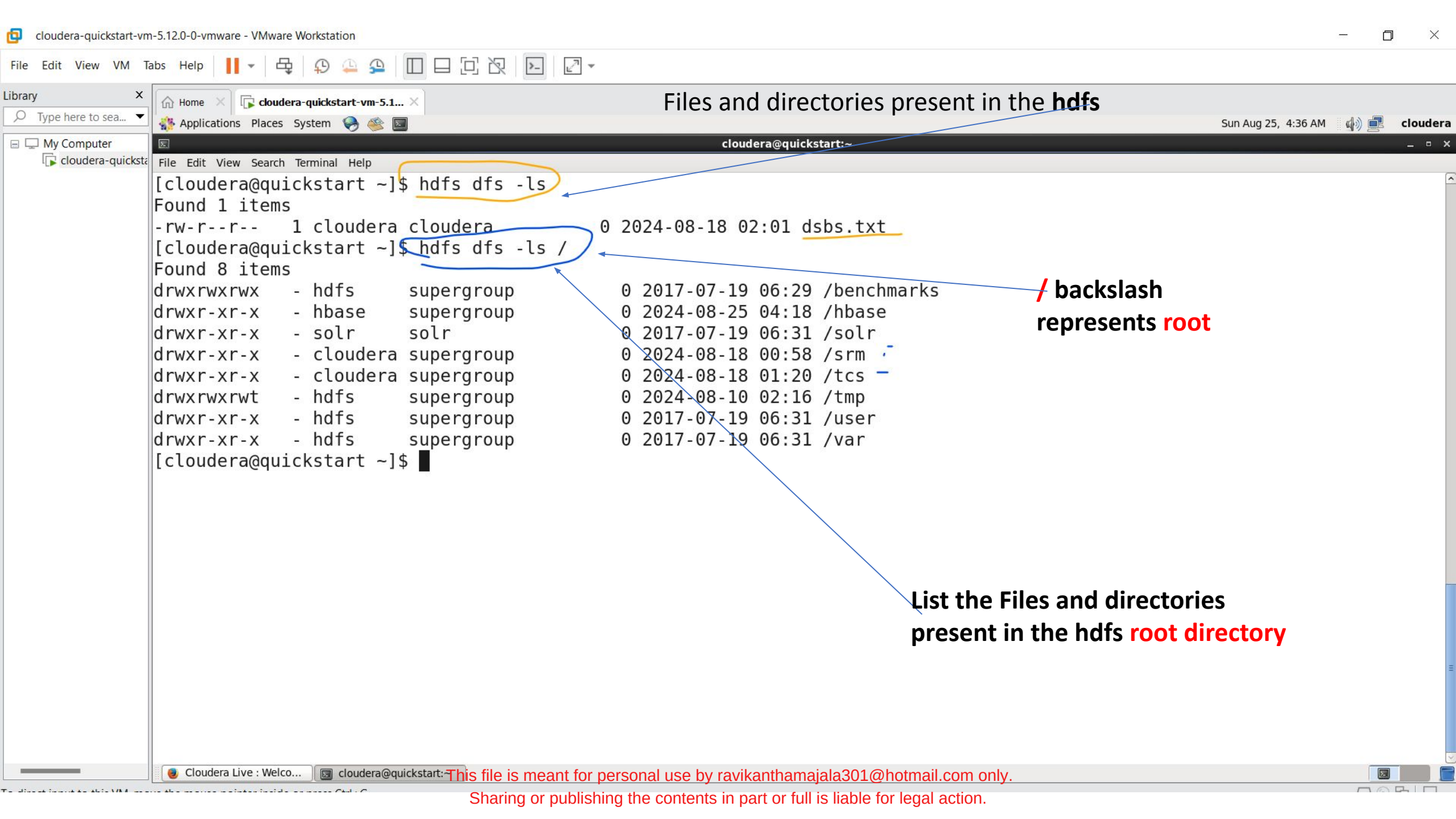
Average block replication: 1.0

Corrupt blocks: 0

Missing replicas: 0 (0.0 %)

Number of data-nodes: 1

It shows the **health status** and various parameters



Files and directories present in the hdfs

```
[cloudera@quickstart ~]$ hdfs dfs -ls
Found 1 items
-rw-r--r--  1 cloudera cloudera 0 2024-08-18 02:01 dsbs.txt
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 8 items
drwxrwxrwx  - hdfs      supergroup 0 2017-07-19 06:29 /benchmarks
drwxr-xr-x  - hbase     supergroup 0 2024-08-25 04:18 /hbase
drwxr-xr-x  - solr      solr      0 2017-07-19 06:31 /solr
drwxr-xr-x  - cloudera  supergroup 0 2024-08-18 00:58 /srm
drwxr-xr-x  - cloudera  supergroup 0 2024-08-18 01:20 /tcs
drwxrwxrwt  - hdfs      supergroup 0 2024-08-10 02:16 /tmp
drwxr-xr-x  - hdfs      supergroup 0 2017-07-19 06:31 /user
drwxr-xr-x  - hdfs      supergroup 0 2017-07-19 06:31 /var
[cloudera@quickstart ~]$
```

/ backslash
represents root

List the Files and directories
present in the hdfs root directory

```
[cloudera@quickstart ~]$ hdfs dfs -mkdir /bigdatatesting
```

```
[cloudera@quickstart ~]$ hdfs dfs -ls /
```

```
Found 9 items
```

drwxrwxrwx	-	hdfs	supergroup	0	2017-07-19	06:29	/benchmarks
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-25	04:47	<u>/bigdatatesting</u>
drwxr-xr-x	-	hbase	supergroup	0	2024-08-25	04:18	/hbase
drwxr-xr-x	-	solr	solr	0	2017-07-19	06:31	/solr
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-18	00:58	/srm
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-18	01:20	/tcs
drwxrwxrwt	-	hdfs	supergroup	0	2024-08-10	02:16	/tmp
drwxr-xr-x	-	hdfs	supergroup	0	2017-07-19	06:31	/user
drwxr-xr-x	-	hdfs	supergroup	0	2017-07-19	06:31	/var

```
[cloudera@quickstart ~]$
```

To create a new **directory** in hdfs
root directory

This is the new
directory we created



```
[cloudera@quickstart ~]$ hdfs dfs -mkdir /bigdatatesting
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 9 items
drwxrwxrwx   - hdfs      supergroup      0 2017-07-19 06:29 /benchmarks
drwxr-xr-x   - cloudera  supergroup      0 2024-08-25 04:47 /bigdatatesting
drwxr-xr-x   - hbase     supergroup      0 2024-08-25 04:18 /hbase
drwxr-xr-x   - solr      solr            0 2017-07-19 06:31 /solr
drwxr-xr-x   - cloudera  supergroup      0 2024-08-18 00:58 /srm
drwxr-xr-x   - cloudera  supergroup      0 2024-08-18 01:20 /tcs
drwxrwxrwt   - hdfs      supergroup      0 2024-08-10 02:16 /tmp
drwxr-xr-x   - hdfs      supergroup      0 2017-07-19 06:31 /user
drwxr-xr-x   - hdfs      supergroup      0 2017-07-19 06:31 /var
[cloudera@quickstart ~]$ hdfs dfs -touchz /bigdatatesting/test.dat
```

test .dat

Is the data file created
in the **new directory** we
created **bigdatatesting**

touchz

This command is used to create a
zero size file

```
[cloudera@quickstart ~]$ hdfs dfs -ls /bigdatatesting
Found 1 items
-rw-r--r-- 1 cloudera supergroup 0 2024-08-25 04:54 /bigdatatesting/test.dat
```

To **list** the file created in the
new directory we created

```
[cloudera@quickstart ~]$ hdfs dfs -touchz /test2.dat
```

To create a new zero size file in the
root directory

```
[cloudera@quickstart ~]$ hdfs dfs -ls /
```

```
Found 10 items
```

drwxrwxrwx	-	hdfs	supergroup	0	2017-07-19	06:29	/benchmarks
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-25	04:54	/bigdatatesting
drwxr-xr-x	-	hbase	supergroup	0	2024-08-25	04:18	/hbase
drwxr-xr-x	-	solr	solr	0	2017-07-19	06:31	/solr
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-18	00:58	/srm
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-18	01:20	/tcs
-rw-r--r--	1	cloudera	supergroup	0	2024-08-25	04:55	<u>/test2.dat</u>
drwxrwxrwt	-	hdfs	supergroup	0	2024-08-10	02:16	/tmp
drwxr-xr-x	-	hdfs	supergroup	0	2017-07-19	06:31	/user
drwxr-xr-x	-	hdfs	supergroup	0	2017-07-19	06:31	/var

This command to display the new file **test2** in the root directory

Create a new file in the **hdfs**

```
[cloudera@quickstart ~]$ hdfs dfs -touchz test3.dat
[cloudera@quickstart ~]$ hdfs dfs -ls
Found 2 items
-rw-r--r--    1 cloudera cloudera    0 2024-08-18 02:01 dsbs.txt
-rw-r--r--    1 cloudera cloudera    0 2024-08-25 04:58 test3.dat
[cloudera@quickstart ~]$
```

List the file in the hdfs

```
cloudera-quickstart-vm-5.12.0-0-vmware - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
cloudera-quickstart
cloudera-quickstart-vm-5.1...
Applications Places System
Sun Aug 25, 5:28 AM
cloudera@quickstart:~
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ hdfs dfs -du -s /bigdatatesting/test.dat
0 0 /bigdatatesting/test.dat
[cloudera@quickstart ~]$ hdfs dfs -du -s /test2.dat
0 0 /test2.dat
[cloudera@quickstart ~]$ hdfs dfs -du -s test3.dat
0 0 test3.dat
[cloudera@quickstart ~]$
```

We have created different
Zero size files in different locations.

To check its size use the command
du – disk usage
s-- size

cloudera-quickstart-vm-5.12.0-0-vmware - VMware Workstation

File Edit View VM Tabs Help

library

My Computer

cloudera-quickstart

Home

cloudera-quickstart-vm-5.1...

Applications Places System

Sun Aug 25, 5:38 AM

cloudera@quickstart:~

```
[cloudera@quickstart ~]$ hdfs dfs -du -s /bigdatatesting/test.dat
0 0 /bigdatatesting/test.dat
[cloudera@quickstart ~]$ hdfs dfs -du -s /test2.dat
0 0 /test2.dat
[cloudera@quickstart ~]$ hdfs dfs -du -s test3.dat
0 0 test3.dat
[cloudera@quickstart ~]$ hdfs dfs -appendToFile - /bigdatatesting/test.dat
hi everyone
welcome to the bigdata course
Bigadat is an amazing platform to handle
huge volume of data
hadoop framework plays a vital role in this
[cloudera@quickstart ~]$ hdfs dfs -du -s /bigdatatesting/test.dat
151 151 /bigdatatesting/test.dat
[cloudera@quickstart ~]$
```

Append content to the file which is present in **hdfs**

You have to enter the values
And then press **ctrl + d**

After entering the data
check the **size**,
Now you can see some
changes in the file size
It shows **151kb**

This command is used to enter some value in the
Newly created files.

```
[cloudera@quickstart ~]$ hdfs dfs -cat /bigdatatesting/test.dat  
hi everyone  
welcome to the bigdata course  
Bigadat is an amazing platform to handle  
huge volume of data  
hadoop framework plays a vital role in this  
[cloudera@quickstart ~]$ █
```

Cat command is used to display the content
in the file test.dat

cloudera-quickstart-vm-5.12.0-0-vmware - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

cloudera-quickstart

Home cloudera-quickstart-vm-5.1...

Applications Places System

Sun Aug 25, 5:57 AM cloudera

cloudera@quickstart:~

File Edit View Search Terminal Help

[cloudera@quickstart ~]\$ cat>>lcnew.dat
hi hello how are you
what's your department name
[cloudera@quickstart ~]\$ ls

cloudera-manager Documents enterprise-deployment.json kerberos lib parcels Templates
cm_api.py Downloads express-deployment.json lcnew.dat mba.txt Pictures Videos
Desktop eclipse hadoop lcnew.dat Music Public workspace

[cloudera@quickstart ~]\$

This command to create a new file in the **local directory**

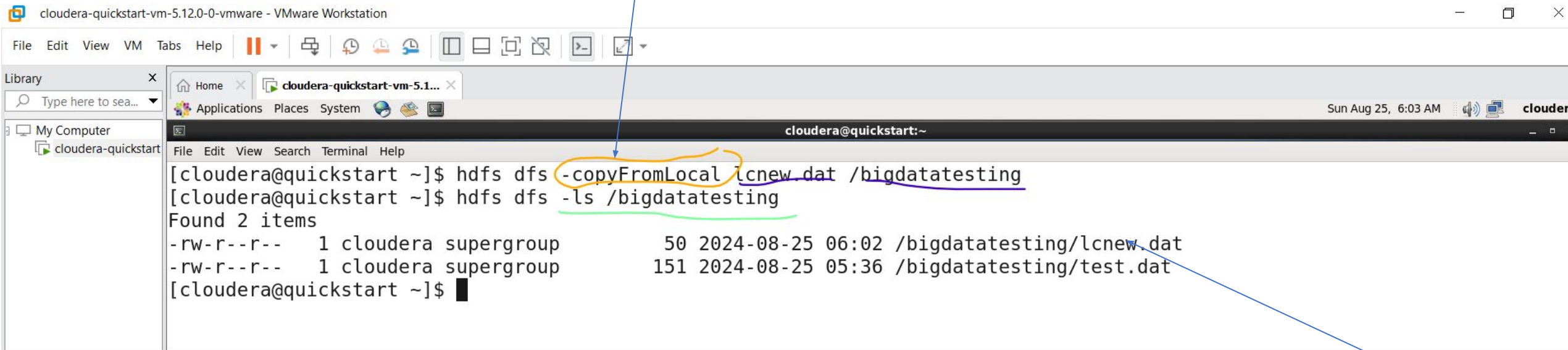
Start entering the content and then press **ctrl + d**

By listing we can see the **file created in the local file system**

copyFromLocal

Command is similar to **put** command

It is used to copy a file from **local directory to hdfs**



```
cloudera-quickstart-vm-5.12.0-0-vmware - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
cloudera-quickstart
cloudera-quickstart-vm-5.1...
Applications Places System
Sun Aug 25, 6:03 AM
cloudera@quickstart:~
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ hdfs dfs -copyFromLocal lcnew.dat /bigdatatesting
[cloudera@quickstart ~]$ hdfs dfs -ls /bigdatatesting
Found 2 items
-rw-r--r-- 1 cloudera supergroup 50 2024-08-25 06:02 /bigdatatesting/lcnew.dat
-rw-r--r-- 1 cloudera supergroup 151 2024-08-25 05:36 /bigdatatesting/test.dat
[cloudera@quickstart ~]$
```

We can see here the new file **lcnew.dat** we created in local directory, is copied to **hdfs** in **bigdatatesting** directory

```
[cloudera@quickstart ~]$ ls
cloudera-manager  Documents  enterprise-deployment.json  kerberos  lib  parcels  Templates
cm_api.py         Downloads  express-deployment.json    lcnew.dat  mba.txt  Pictures  Videos
Desktop          eclipse    hadoop                     lctest.dat Music    Public    workspace
[cloudera@quickstart ~]$
```

Can you see there is a file **lctest.dat** in the Local file directory

Using put command also we can **put** a file from local system to hdfs directories

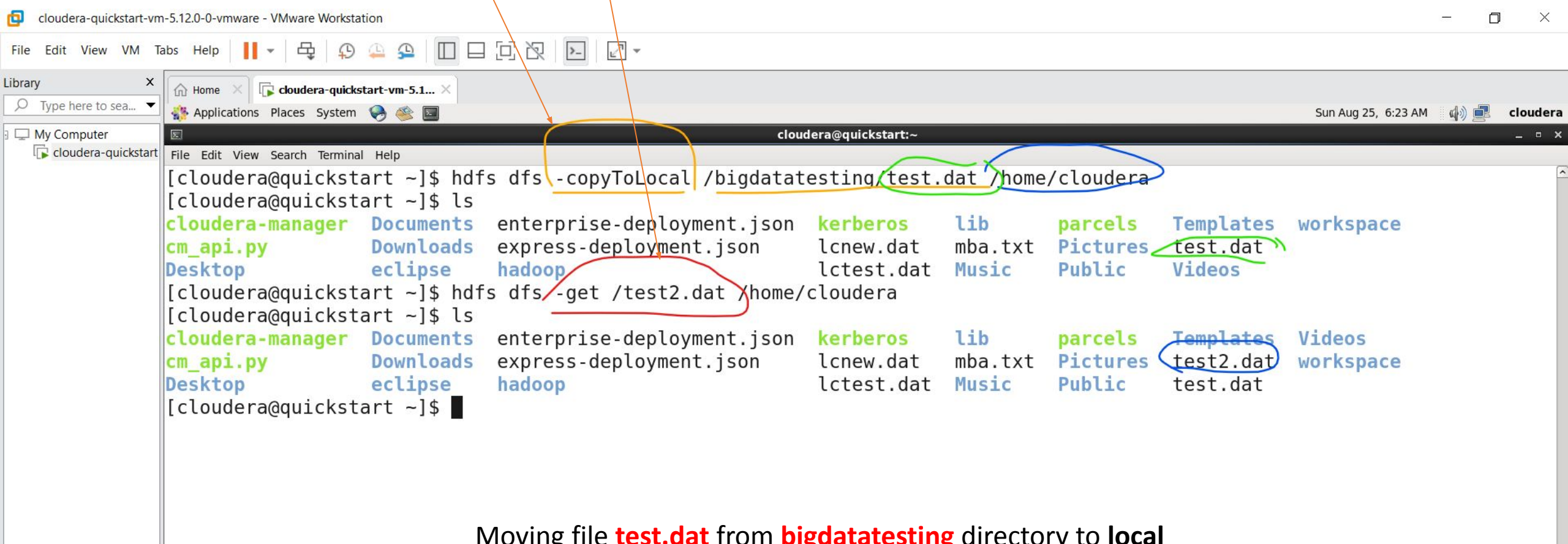
```
[cloudera@quickstart ~]$ hdfs dfs -put lctest.dat /bigdatatesting
[cloudera@quickstart ~]$ hdfs dfs -ls /bigdatatesting
Found 3 items
-rw-r--r--  1 cloudera supergroup      50 2024-08-25 06:02 /bigdatatesting/lcnew.dat
-rw-r--r--  1 cloudera supergroup      86 2024-08-25 06:11 /bigdatatesting/lctest.dat
-rw-r--r--  1 cloudera supergroup    151 2024-08-25 05:36 /bigdatatesting/test.dat
[cloudera@quickstart ~]$
```

Now you can see **lctest.dat** is moved to the **hdfs** in **bigdatatesting** directory.

Copy file from hdfs to local directory

copyToLocal and **get**

Both the commands work in the same way



The screenshot shows a terminal window titled 'cloudera-quickstart-vm-5.12.0-0-vmware - VMware Workstation'. The terminal prompt is 'cloudera@quickstart:~'. The first command executed is 'hdfs dfs -copyToLocal /bigdatatesting/test.dat /home/cloudera', with 'copyToLocal' circled in yellow and the source path circled in green. The second command is 'ls', showing a directory listing with files like 'cloudera-manager', 'cm_api.py', 'Desktop', 'Documents', 'Downloads', 'eclipse', 'enterprise-deployment.json', 'express-deployment.json', 'hadoop', 'kerberos', 'lcnw.dat', 'lctest.dat', 'lib', 'mba.txt', 'Music', 'parcels', 'Pictures', 'Public', 'Templates', 'Videos', and 'workspace'. The file 'test.dat' is circled in green. The third command is 'hdfs dfs -get /test2.dat /home/cloudera', with 'get' circled in red. The fourth command is 'ls', showing a directory listing with files like 'cloudera-manager', 'cm_api.py', 'Desktop', 'Documents', 'Downloads', 'eclipse', 'enterprise-deployment.json', 'express-deployment.json', 'hadoop', 'kerberos', 'lcnw.dat', 'lctest.dat', 'lib', 'mba.txt', 'Music', 'parcels', 'Pictures', 'Public', 'Templates', 'Videos', and 'workspace'. The file 'test2.dat' is circled in blue.

```
cloudera@quickstart:~$ hdfs dfs -copyToLocal /bigdatatesting/test.dat /home/cloudera
cloudera@quickstart:~$ ls
cloudera-manager Documents enterprise-deployment.json kerberos lib parcels Templates workspace
cm_api.py Downloads express-deployment.json lcnw.dat mba.txt Pictures test.dat
Desktop eclipse hadoop lctest.dat Music Public Videos
cloudera@quickstart:~$ hdfs dfs -get /test2.dat /home/cloudera
cloudera@quickstart:~$ ls
cloudera-manager Documents enterprise-deployment.json kerberos lib parcels Templates Videos
cm_api.py Downloads express-deployment.json lcnw.dat mba.txt Pictures test2.dat workspace
Desktop eclipse hadoop lctest.dat Music Public test.dat
cloudera@quickstart:~$
```

Moving file **test.dat** from **bigdatatesting** directory to local directory

Moving file **test2.dat** from **root directory** to local directory

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Sharing or publishing the contents in part or full is liable for legal action.


```
[cloudera@quickstart ~]$ hdfs dfs -mkdir /hadooptesting
```

Let's create a new directory in the name **hadooptesting**

```
[cloudera@quickstart ~]$ hdfs dfs -ls /
```

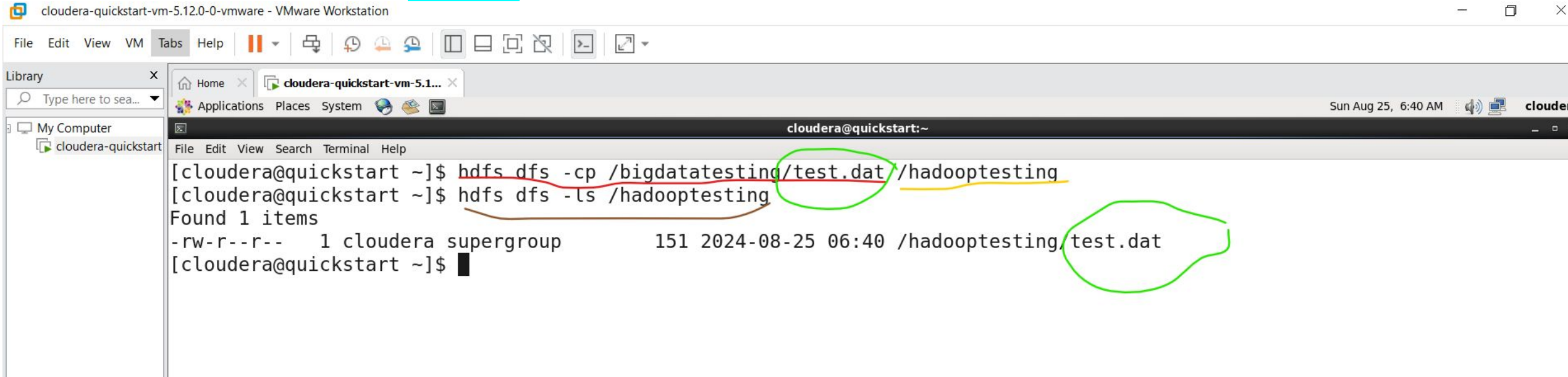
Found 11 items

drwxrwxrwx	-	hdfs	supergroup	0	2017-07-19	06:29	/benchmarks
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-25	06:11	/bigdatatesting
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-25	06:34	/hadooptesting
drwxr-xr-x	-	hbase	supergroup	0	2024-08-25	04:18	/hbase
drwxr-xr-x	-	solr	solr	0	2017-07-19	06:31	/solr
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-18	00:58	/srm
drwxr-xr-x	-	cloudera	supergroup	0	2024-08-18	01:20	/tcs
-rw-r--r--	1	cloudera	supergroup	0	2024-08-25	04:55	/test2.dat
drwxrwxrwt	-	hdfs	supergroup	0	2024-08-10	02:16	/tmp
drwxr-xr-x	-	hdfs	supergroup	0	2017-07-19	06:31	/user
drwxr-xr-x	-	hdfs	supergroup	0	2017-07-19	06:31	/var

We can see a new file created in the hdfs root directory in the name **hadooptesting**

Cp is the command
for copy

We copied the file **test.dat** from
bigdatatesting directory to **hadooptesting
directory**



cloudera-quickstart-vm-5.12.0-0-vmware - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

cloudera-quickstart

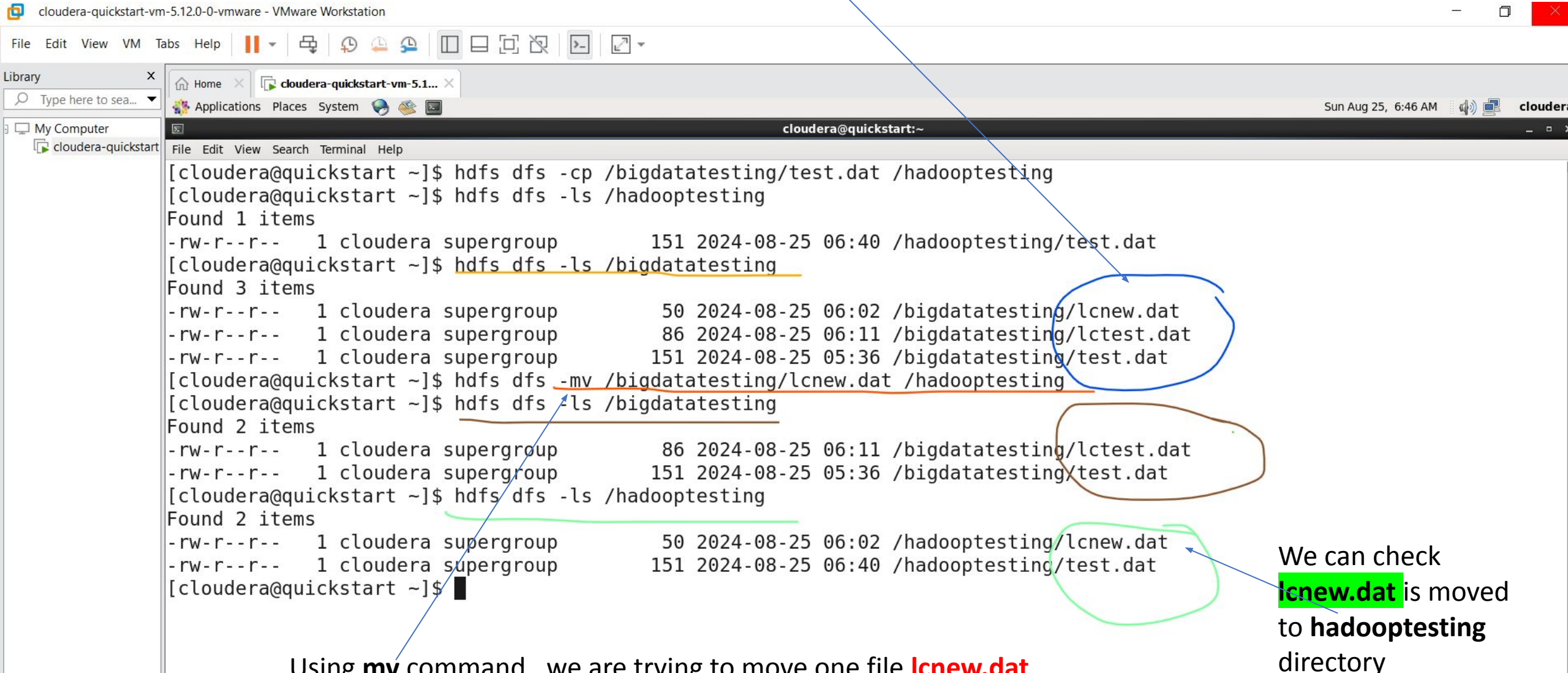
cloudera-quickstart-vm-5.1...

Applications Places System

cloudera@quickstart:~

```
[cloudera@quickstart ~]$ hdfs dfs -cp /bigdatatesting/test.dat /hadooptesting
[cloudera@quickstart ~]$ hdfs dfs -ls /hadooptesting
Found 1 items
-rw-r--r--  1 cloudera supergroup      151 2024-08-25 06:40 /hadooptesting/test.dat
[cloudera@quickstart ~]$
```

Initially in **bigdatatesting** directory there are 3 files.



```
cloudera-quickstart-vm-5.12.0-0-vmware - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
cloudera-quickstart
cloudera@quickstart:~
[cloudera@quickstart ~]$ hdfs dfs -cp /bigdatatesting/test.dat /hadooptesting
[cloudera@quickstart ~]$ hdfs dfs -ls /hadooptesting
Found 1 items
-rw-r--r-- 1 cloudera supergroup 151 2024-08-25 06:40 /hadooptesting/test.dat
[cloudera@quickstart ~]$ hdfs dfs -ls /bigdatatesting
Found 3 items
-rw-r--r-- 1 cloudera supergroup 50 2024-08-25 06:02 /bigdatatesting/lcnew.dat
-rw-r--r-- 1 cloudera supergroup 86 2024-08-25 06:11 /bigdatatesting/lctest.dat
-rw-r--r-- 1 cloudera supergroup 151 2024-08-25 05:36 /bigdatatesting/test.dat
[cloudera@quickstart ~]$ hdfs dfs -mv /bigdatatesting/lcnew.dat /hadooptesting
[cloudera@quickstart ~]$ hdfs dfs -ls /bigdatatesting
Found 2 items
-rw-r--r-- 1 cloudera supergroup 86 2024-08-25 06:11 /bigdatatesting/lctest.dat
-rw-r--r-- 1 cloudera supergroup 151 2024-08-25 05:36 /bigdatatesting/test.dat
[cloudera@quickstart ~]$ hdfs dfs -ls /hadooptesting
Found 2 items
-rw-r--r-- 1 cloudera supergroup 50 2024-08-25 06:02 /hadooptesting/lcnew.dat
-rw-r--r-- 1 cloudera supergroup 151 2024-08-25 06:40 /hadooptesting/test.dat
[cloudera@quickstart ~]$
```

Using **mv** command , we are trying to move one file **lcnew.dat** from **bigdatatesting** directory to **Hadoop testing** directory

We can check **lcnew.dat** is moved to **hadooptesting** directory

rm command is used to remove a **file** from the directory

```
[cloudera@quickstart ~]$ hdfs dfs -rm /bigdatatesting/test.dat
Deleted /bigdatatesting/test.dat
[cloudera@quickstart ~]$ hdfs dfs -ls /bigdatatesting
Found 1 items
-rw-r--r--  1 cloudera supergroup      86 2024-08-25 06:11 /bigdatatesting/lctest.dat
[cloudera@quickstart ~]$
```

Here the file **test.dat** is removed from the **bigdatatesting** directory

Using **rmdir**

We can remove a directory , **but it won't remove if the directory contains data in it**

```
[cloudera@quickstart ~]$ hdfs dfs -rmdir /bigdatatesting
rmdir: `/bigdatatesting': Directory is not empty
[cloudera@quickstart ~]$ hdfs dfs -rm -r /bigdatatesting
Deleted /bigdatatesting
[cloudera@quickstart ~]$ █
```

Using **-rm -r**

We can remove a directory , **even it has data in it .**